

A sphere inside a point, while you are still walking on it

A faithful reconstruction of the reduced-sphere paper in nonstandard analysis, with one new scale-loss theorem and its spatial-computing consequences.

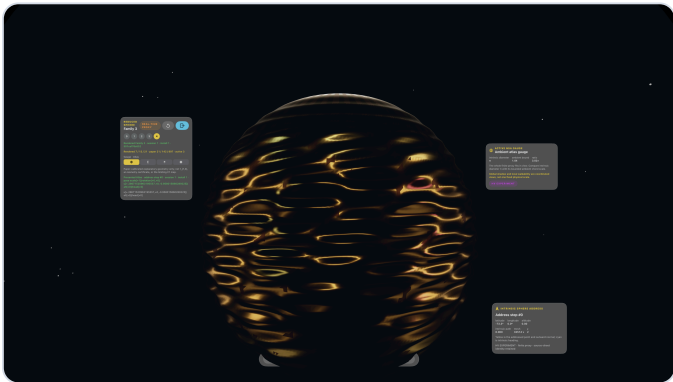
The address is the geometry. The rendered point is only the current observation of it.

Canonical source: [report-source.md](#)

29 AUGUST 2026 · UNOFFICIAL RESEARCH COMPANION

One address. Four honest views.

These are generated by Hevea Vision on a finite REAL-TIME PROXY mesh. They visualize the model; they do not certify the paper's limiting isometry or physical-headset comfort. Habitat and Hover retain their semantic HUD receipts, while the underfoot surface falls below the simulator's forward capture.



Atlas • the reduced ball as an object



Habitat • foot-scale controls; underfoot shell below frame



Hover • signed-altitude controls; near-field shell below frame



Interior • the containing-ball gauge

A faithful reconstruction of Hévéa's reduced sphere in nonstandard analysis

Reader: Alok Singh

Status: canonical research source, 2026-08-29

Companion app: [Hevea Vision](#)

Tracked work: [ALOK-846](#)

This document is intentionally optimized for one reader. It assumes that you prefer theorem-shaped claims, typed boundaries, inspectable dependencies, and an honest separation between a proof, a finite computation, and a visual metaphor. It does not assume prior fluency with convex integration. It does assume comfort with derivatives, Riemannian metrics, compactness, transfer, and the idea that a limit can be represented by an unlimited hyperfinite stage.

The exposition is newly written. It follows the mathematical order and equation numbering of Bartzos-Borrelli-Denis-Lazarus-Rohmer-Thibert without reproducing their copyrighted prose or figures.

Claim labels

- **PAPER** - stated or proved in the reduced-sphere paper.
 - **NSA-EQUIV** - an equivalent reformulation by transfer, hyperfinite indexing, or standard part.
 - **NEW-COROLLARY** - a result derived here by combining the paper with standard nonstandard-analysis machinery.
 - **APP-MODEL** - a product/visualization consequence, not a theorem about the paper's surface.
 - **OPEN** - a plausible research direction for which this report does not supply a proof.
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0. The whole idea in ninety seconds

The ordinary unit sphere has intrinsic diameter π . A smooth sphere is rigid at C^2 regularity, but Nash-Kuiper flexibility begins at C^1 . Hévéa turns that existence phenomenon into an explicit boundary-aware convex integration scheme.

PAPER For every ordinary radius $0 < r < 1$, there is a C^1 isometric map of the round unit sphere into the Euclidean ball $B_r \subset \mathbb{R}^3$. The construction keeps two translated round caps smooth and corrugates the equatorial belt. Three primitive metric defects are corrected in succession on nested ribbons; repeating the three-step stage gives a C^1 -Cauchy sequence whose limit has exactly the round metric.

NSA-EQUIV Choose an unlimited hyperinteger H and look at the transferred stage $*f_{H,3}$. Its metric defect is infinitesimal, all unlimited stages are infinitesimally close in C^1 , and its C^1 shadow is the paper's limit f_∞ .

NEW-COROLLARY Transfer the final theorem itself to a positive infinitesimal radius ε . Then an internal exact isometry

$$F_\varepsilon : {}^*S^2 \longrightarrow {}^*B_\varepsilon$$

exists. Every image point is infinitesimally close to the origin, so the pointwise standard shadow is constant. Nevertheless every standard path keeps its nonzero intrinsic length internally. Standard part therefore destroys the global picture before it preserves the walkable geometry. The derivative must rotate on scales smaller than every standard observational scale.

That is the app's architecture in one sentence:

Keep the visitor's address on the intrinsic source sphere; render a chosen extrinsic gauge around that address.

The sphere has no infinite minimal geodesic distances. The quantity that can become unlimited is the ratio

$$\frac{\text{intrinsic separation}}{\text{ambient chord separation}}.$$

1. Seven things worth retaining

1. The famous rendered sphere is a finite $f_{1,3}$ mesh, not f_∞ .
2. Its three reported visible ridge counts are 21, 142, 997; those are not three iterations of the whole convergence proof, but the three primitive corrections inside the first stage.
3. The paper's small target-metric error 0.03 at the final displayed row is not a global round-sphere error. Its reported average error against the round metric is still 0.66.
4. Boundary conditions are the intellectual center: the active ribbons expand toward the caps while every finite stage glues back to the unchanged map.
5. “ N is unlimited” never licenses dropping $O(C/N)$ until the hidden constant C is controlled relative to N .
6. At standard radius, an unlimited construction stage is C^1 -nearstandard and shadows to the isometry. At infinitesimal radius, the transferred exact isometry has a constant pointwise shadow and is not C^1 -nearstandard.
7. Spatial proximity is not intrinsic identity. Locomotion and collision must remember the parameter-sheet address.

2. Dependency spine

flowchart LR

```
A[Strictly short initial ribbon f0] -->|positive primitive defects| B[One curve corrugation]
B -->|phase-shifted flow family| C[Correct primitive 1]
C --> D[Correct primitive 2]
D --> E[Correct primitive 3]
E -->|nested ribbons and target slack| F[One stage fk,3]
F -->|summable C1 changes| G[C1 limit f infinity]
G -->|metric defect tends to zero| H[Round metric exactly]
F -->|unlimited H| I[Hyperfinite micro-isometry]
I -->|C1 standard part, standard r| G
H -->|transfer final theorem to infinitesimal epsilon| J[Exact internal isometry in B epsilon]
J -->|pointwise standard part| K[Constant global shadow]
J -->|internal pullback metric| L[Nonzero intrinsic path lengths]
```

The proof is not “add wrinkles until it looks right.” It is a controlled induction maintaining five obligations at once:

- quotient periodicity;
- positivity of the next primitive defect;
- cap/ribbon boundary attachment;
- decay of metric error; and
- summability of derivative changes.

3. Notation and the geometric type signature

Let

$$h(x, y) = (\cos y \cos x, \cos y \sin x, \sin y)$$

be the usual parametrization of the unit sphere away from the two polar caps. The belt domain is

$$D = S^1 \times (-y_\infty, y_\infty), \quad S^1 = \mathbb{R}/2\pi\mathbb{Z}.$$

The round first fundamental form is

$$I_h = \begin{pmatrix} \cos^2 y & 0 \\ 0 & 1 \end{pmatrix}.$$

For a map $f : D \rightarrow \mathbb{R}^3$, write $I_f = f^*\langle -, - \rangle$. The isometry equation is the nonlinear first-order PDE

$$I_f = I_h.$$

The paper fixes three linear forms

$$\ell_1(X, Y) = \frac{X+Y}{\sqrt{2}}, \quad \ell_2(X, Y) = \frac{Y-X}{\sqrt{2}}, \quad \ell_3(X, Y) = Y. \quad (2.1)$$

Every symmetric 2×2 form B decomposes uniquely as

$$B = \sum_{i=1}^3 \rho_i(B) \ell_i \otimes \ell_i. \quad (2.2)$$

For $B = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$, this convention gives

$$\rho_1(B) = E + F, \quad \rho_2(B) = E - F, \quad \rho_3(B) = G - E.$$

This basis is the algorithmic hinge. Each corrugation step targets exactly one ρ_i , while controlling what happens in $\ker \ell_i$.

The theorem-shaped program

The construction has the following conceptual type:

```
input:
  radius r with 0 < r < 1
  strictly short embedded initial map f0 inside B_r
  positive primitive defect coordinates

output:
  C1 map f∞ : S2 -> B_r
  pullback metric certificate I_f∞ = I_h
  exact attachment to two translated round caps

construction evidence:
  recursively selected domains, metrics, frequencies, stage maps
  C0 perturbation budget
  summable C1 increments
  metric-defect convergence
```

The paper repeatedly calls the result an embedding, while its displayed Corollary 10 formally concludes “isometric map.” Metric equality gives a local C^1 isometric immersion. The extracted argument does not isolate global injectivity in a separate named lemma. This report therefore says “isometric map” at the theorem boundary and treats global embedding as the authors’ broader stated claim.

Part I - The paper, faithfully reconstructed

4. The initial map: shorten first, repair later

[PAPER, Sections 2-3] Split the sphere into closed north/south caps and an open equatorial belt. Translate the two caps toward the origin. On the belt, seek a rotational map

$$f_0(x, y) = (X(y) \cos x, X(y) \sin x, Z(y)).$$

It must satisfy four conditions:

- **(C1)** smoothness on the open belt;
- **(C2)** image containment in B_r ;
- **(C3)** C^1 attachment to the translated caps;
- **(C4)** strict positivity of every primitive coordinate $\rho_i(I_h - I_{f_0})$.

The last condition says more than ordinary shortness, but it implies shortness. The missing metric lies in the open positive cone spanned by the three fixed primitive forms. This fixed cone is what lets the same three directions survive the entire induction.

For the rotational map,

$$I_h - I_{f_0} = \begin{pmatrix} \cos^2 y - X(y)^2 & 0 \\ 0 & 1 - X'(y)^2 - Z'(y)^2 \end{pmatrix}.$$

The boundary values at $\pm y_\infty$ match translated round caps. If the translation size is η , then, for example, the north endpoint obeys

$$X(y_\infty) = \cos y_\infty, \quad Z(y_\infty) = \sin y_\infty - \eta,$$

$$X'(y_\infty) = -\sin y_\infty, \quad Z'(y_\infty) = \cos y_\infty. \quad (3.7)$$

The south data are symmetric. The proof of Proposition 1 builds a piecewise profile, uses one-dimensional convex integration on its middle formal solution, and smooths without crossing the strict inequalities. The numerical implementation described by the authors used a Hermite cubic spline reparametrized by another Hermite cubic spline, producing degree-nine coordinates. Those coefficients are not printed.

What is exact in the app

The round parametrization, translated caps, endpoint equations, and primitive forms are source formulas. Any interactive central profile is a reconstruction unless the unpublished coefficients are obtained from the authors. The app's current profile is therefore labelled **REAL-TIME PROXY**, even when its seam equations pass numerically.

5. One corrugated curve

[PAPER, Section 4] Let γ be a source curve. Suppose its current image speed is no larger than a target speed $r(s)$. Write t for the unit tangent of the image curve and n for the surface normal. The corrected curve is the integral of a vector of norm r :

$$\Gamma(s) = f_0(\gamma(0)) + \int_0^s r(\sigma)(\cos \theta t + \sin \theta n) d\sigma. \quad (4.8)$$

The phase is

$$\theta(\sigma) = \alpha(\sigma) \cos(2\pi N\sigma),$$

where

$$\alpha(\sigma) = J_0^{-1} \left(\frac{\|(f_0 \circ \gamma)'(\sigma)\|}{r(\sigma)} \right). \quad (4.9)$$

The first monotone branch of J_0 is used, up to its first positive zero. The Bessel inverse is not ornament: averaging the rotated unit vector over one phase cycle returns the old tangent component, while its pointwise norm is the larger target speed. High frequency changes position by only $O(1/N)$, even though the derivative rotates appreciably.

This is the geometric paradox in miniature:

small positional motion + large derivative motion = hidden extra length

6. A family of curves with boundary-compatible phase

A surface needs a one-parameter family of source curves. These are flow lines of a nonhorizontal field w on the lifted cylinder $\tilde{D}$. A naive common initial phase makes the wrong ridge direction and fails to correct the horizontal metric component. The paper instead gives each flow line an affine starting parameter a_p and integrates from its south-boundary point:

$$\tilde{F} \circ \gamma_p(s) = f_0([p]) + \int_{a_p}^s r(\gamma_p(\sigma)) (\cos \theta t + \sin \theta n) d\sigma. \quad (4.10)$$

Two obligations appear.

First, the lift must descend through $x \sim x + 2\pi$. If

$$\Delta = a_{p+2\pi e_X} - a_p,$$

then Lemma 2 requires

$$N\Delta \in \mathbb{Z}.$$

Second, a transverse derivative must remain close. With $\nu = \partial\gamma/\partial x$, Lemma 3 gives

$$\|\nu \cdot F - \nu \cdot f_0\|_\infty = O(1/N).$$

The arithmetic periodicity condition is exact. It may not be weakened to “almost integral,” either classically or nonstandardly.

7. Correct exactly one primitive coordinate

[PAPER, Section 5] At correction i , let

$$\rho_i = \rho_i(I_h - I_{f_{i-1}}).$$

Define the intermediate metric

$$\mu_i = I_{f_{i-1}} + \rho_i \ell_i \otimes \ell_i. \quad (5.14)$$

Let v_i span $\ker \ell_i$:

$$v_1 = \frac{(-1, 1)}{\sqrt{2}}, \quad v_2 = -\frac{(1, 1)}{\sqrt{2}}, \quad v_3 = (-1, 0). \quad (5.15)$$

Choose w_i by

$$\ell_i(w_i) = 1, \quad \text{quad} \mu_i(w_i, v_i) = 0.$$

The target speed is

$$r^2 = \|df_{i-1}(w_i)\|^2 + \rho_i. \quad (5.13)$$

Lemma 4 forces the phase origin

$$a_p = \ell_i(p - (0, 0)) + \text{constant}.$$

Consequently the ridge lines are parallel to v_i . Lemmas 5-6 show that the v_i derivative changes by only $O(1/N_i)$, the induced metric is $O(1/N_i)$ -close to μ_i , and the selected primitive defect becomes $O(1/N_i)$.

One step does not make the surface isometric. It corrects one coordinate while perturbing the others by a controlled amount.

8. Why nested ribbons are necessary

[PAPER, Section 6] Boundary gluing perturbs the metric, and the available primitive margins vanish toward the cap seams. The paper introduces two kinds of slack at once.

First, target metrics approach the round metric gradually:

$$I_k = \tau_k^2 I_{f_0} + (1 - \tau_k^2) I_h, \quad \tau_0 = 1, \quad \tau_k \downarrow 0.$$

Second, three nested ribbons per stage expand toward the cap boundaries:

$$D_{1,1} \subset D_{1,2} \subset D_{1,3} \subset D_{2,1} \subset \cdots \subset D.$$

The missing boundary strips are selected so that

$$\|I_h - I_{f_0}\| \leq \tau_k^2 \quad \text{outside } D_{k,3}.$$

Altered targets $I_{k,i}$ interpolate between I_{f_0} near the inactive boundary and I_k on the smaller active ribbon. They are monotone:

$$I_{f_0} \leq I_{k,i} \leq I_{\ell,j} \leq I_h \quad (6.18)$$

in lexicographic stage order.

The raw corrugated map $F_{k,i}$ is blended back to the previous map:

$$f_{k,i} = (1 - \chi_{k,i})F_{k,i} + \chi_{k,i}f_{k,i-1}. \quad (6.20)$$

The blend is arranged so that $f_{k,i} = F_{k,i}$ on the smaller ribbon and $f_{k,i} = f_0$ outside the current ribbon. Thus every finite stage remains C^1 -attached to the translated caps.

The quotient requirement becomes

$$2\pi N_{k,i} \ell_i(e_X) \in \mathbb{Z}. \quad (\text{Lemma 7})$$

9. The Stage and Step theorems

[PAPER, Sections 7-8] Lemma 8 bounds the gluing difference in C^1 by $O(1/N_{k,i})$.

The Step theorem packages one correction. With

$$D_{k,i} = I_{k,i} - I_{f_{k,i-1}},$$

and positive selected primitive coordinate, it gives constants independent of the chosen frequency such that

$$\|f_{k,i} - f_{k,i-1}\|_\infty = O(1/N_{k,i}),$$

$$\|df_{k,i} - df_{k,i-1}\|_\infty \leq \frac{C_{k,i}}{N_{k,i}} + 2\sqrt{7} \|\rho_i(D_{k,i})\|^{1/2},$$

$$\|\mu_{k,i} - I_{f_{k,i}}\|_\infty \leq \frac{C'_{k,i}}{N_{k,i}}. \quad (\text{Theorem 11})$$

The Stage theorem packages three corrections. Frequencies can be chosen so that positivity propagates to the next stage and

$$\|I_{k,3} - I_{f_{k,3}}\|_\infty \leq C\sqrt{\tau_{k-1}},$$

$$\|df_{k,3} - df_{k-1,3}\|_\infty \leq C'\tau_{k-1}, \quad (\text{Theorem 9})$$

where C, C' are independent of k .

Lemma 12 maintains that each characteristic field is nonhorizontal, so its flow lines really cross the ribbon. This is qualitative nonhorizontality; it does not promise a stage-independent lower bound on the crossing angle.

10. Convergence

[PAPER, Corollary 10] If

$$\sum_{k=0}^{\infty} \tau_k < \infty,$$

then the derivative increments are summable. The maps agree on the boundary, so $(f_{k,3})$ converges in C^1 to a map f_∞ . The metric estimate and $I_k \rightarrow I_h$ imply

$$I_{f_\infty} = I_h.$$

The C^0 perturbations can be chosen with any prescribed summable budget, so f_∞ stays as close to f_0 as necessary. Choosing that budget smaller than the initial image's clearance from ∂B_r preserves ball containment.

This is the exact isometry. No finite mesh receives that claim.

11. The C^1 -fractal expansion

[PAPER, Section 9] One corrugation rotates an adapted tangent-normal frame by a matrix $C_{k,i} \in SO(3)$. The limit Gauss map is expressed through an ordered tail product. On the band first activated at (k,i) , Theorem 13 uses

$$R(k,i) = \left(\prod_{\ell=k+1}^\infty \prod_{j=1}^3 C_{\ell,j} \right) \left(\prod_{j=i}^3 C_{k,j} \right).$$

Near the equator the normal sees the full product from the earliest stage. Moving toward a cap replaces that product by later and later remainders. In the cap-to-equator direction, new corrugation families appear over successively larger central subsegments and with visibly larger amplitude. This is what the authors call a C^1 -fractal expansion.

This terminology describes a generalized Riesz-product structure in the Gauss map. It is not, by itself, a Hausdorff-dimension theorem.

12. What the published computation actually shows

[PAPER, pp. 15-16] The gallery's central object is a discretization of the first stage after three primitive corrections, $f_{1,3}$.

row	corrugation number $N_{k,j}$	visible ridges	average/max error vs. round metric	average/max error vs. current altered target
initial	-	-	0.90 / 1.17	-
(1,1)	4.72	21	0.83 / 1.03	0.14 / 0.24
(1,2)	31.96	142	0.73 / 0.95	0.07 / 0.16
(1,3)	334.92	997	0.66 / 0.94	0.03 / 0.18

The run used a 4000×20000 grid, a 16-core CPU, 64 GB RAM, and about 2 hours 46 minutes. The smaller last column is often misread. Its target metric is altered and equals the intended stage target only on the relevant subdomain. The paper itself warns that the maximum comparison there is not globally meaningful.

The authors chose a factor-two visual reduction. Their theorem permits every positive standard radius.

The user's image

The supplied 533×300 reference image is a compressed derivative of the official 7680×4320 first “Sur la sphère / On the sphere” gallery frame:

https://hevea-project.fr/imgSphere/070_02.jpg

It is not merely mood art. It is direct evidence that a first-person terrain view belongs to the original researchers' own visual program. The new app turns that static viewpoint into a navigable intrinsic address.

Part II - The nonstandard reconstruction

13. Choose one foundation and say which one

The main development uses a sufficiently rich superstructure enlargement

$$* : V(\mathbb{R}) \longrightarrow V(*\mathbb{R})$$

with transfer, infinite hyperintegers, and standard part for limited finite-dimensional values and nearstandard points.

This is a presentation choice, not a mathematical claim that enlargement semantics are superior to Internal Set Theory. It matches the paper's indexed families and lets us write $*f_H$, internal hyperfinite products, and transferred integrals without syntactic strain.

Nelson's IST can express the same reasoning inside an enriched language for ordinary set theory. Its three axiom schemes are Transfer, Idealization, and Standardization. In IST, "standard" is a predicate; there is no syntactically separate set called $*\mathbb{R}$. One must not mix IST quantifiers and star-map notation without a translation paragraph.

Two safety rules are non-negotiable:

- 1. $x \approx y$ is external. It cannot occur inside a formula to which transfer is applied.
- 2. The collection of all infinitesimals is external. Treating it as an internal set is illegal set formation.

14. Dictionary

Classical object	Nonstandard object	Required caution
limit stage f_∞	$*f_{H,3}$ for unlimited H	only if the standard sequence was selected first or a saturation argument constructs it
error tends to zero	error is infinitesimal at every unlimited index	hidden constants must be dominated
Cauchy in C^1	unlimited stages are mutually infinitesimally close in C^1	boundedness in an infinite-dimensional Banach space is not nearstandardness
metric equality in the limit	micro-isometry plus C^1 -nearstandardness	pointwise shadow alone cannot commute with d
infinite ordered product	internal hyperfinite ordered product	order matters; shadow independence requires convergence
increasingly fine mesh	hyperfinite partition	it must resolve the active frequency, not merely have infinitely many cells
near boundary	a point in the boundary monad	a positive primitive margin may be infinitesimal

Define a micro-isometric internal map by

$$\|*I_h - I_F\|_\infty \approx 0.$$

Define C^1 -nearstandardness by the existence of a standard $f \in C^1(S^2, \mathbb{R}^3)$ with

$$\|F - *f\|_\infty + \|dF - *df\|_\infty \approx 0.$$

The derivative term is the gate that distinguishes the two radius regimes below.

15. Hyperfinite shadow theorem at a standard radius

Fix standard data:

- $0 < r < 1$;
- an initial map f_0 ;
- a summable sequence (τ_k) ;
- one recursively selected standard sequence of all $D_{k,i}, I_{k,i}, N_{k,i}, F_{k,i}, f_{k,i}$.

Ordinary dependent choice is enough to select that sequence from the paper's stage-by-stage existence statements. Extend the completed sequence by $*$. Let $H \in {}^*\mathbb{N} \setminus \mathbb{N}$ and set

$$\mathcal{F}_H = *f_{H,3}.$$

Theorem 15.1 - hyperfinite shadow NSA-EQUIV

For every unlimited H :

1. $\mathcal{F}_H$ is an internal $*C^1$ map on $*S^2$, internally smooth on ribbon and cap interiors, with transferred exact C^1 cap attachment.
2. $\|*I_h - I_{\mathcal{F}_H}\|_\infty \approx 0$.
3. For unlimited H, K , $\|\mathcal{F}_H - \mathcal{F}_K\|_{C^1} \approx 0$.
4. There is a unique standard $f_\infty \in C^1$ such that $\mathcal{F}_H \approx_{C^1} *f_\infty$.
5. $I_{f_\infty} = I_h$.

Proof spine

Item 1 refers to the transferred construction at hyperfinite index H . The paper does not claim that all higher derivatives match across the cap seams.

The metric estimate splits as

$$\|*I_h - I_{\mathcal{F}_H}\|_\infty \leq \|*I_h - *I_{H,3}\|_\infty + C\sqrt{{}^*\tau_{H-1}}.$$

The second term is infinitesimal. For the first, write the altered target as

$$I_{k,3} = (1 - \lambda_{k,3})I_{f_0} + \lambda_{k,3}I_k.$$

Then

$$I_h - I_{k,3} = (1 - \lambda_{k,3})(I_h - I_{f_0}) + \lambda_{k,3}(I_h - I_k).$$

On $D_{k,2}$, $\lambda_{k,3} = 1$, so the remaining target error is bounded by $\tau_k^2 \|I_h - I_{f_0}\|_\infty$. Outside $D_{k,2}$, the first term tends uniformly to zero because $D_{k,2} \uparrow D$ and the continuous defect $I_h - I_{f_0}$ vanishes on ∂D . Thus $I_{k,3} \rightarrow I_h$ uniformly, and transfer makes the first term infinitesimal at unlimited H .

For $H > K$, telescope the derivative:

$$\|d\mathcal{F}_H - d\mathcal{F}_K\|_\infty \leq C' \sum_{j=K+1}^H {}^*\tau_{j-1}.$$

Every hyperfinite tail beyond an unlimited K is infinitesimal because the standard series is summable. The analogous C^0 increments were selected with a summable budget. These standard estimates first show that the standard sequence $(f_{k,3})$ is Cauchy in the standard Banach space $C^1(S^2, \mathbb{R}^3)$. Standard completeness gives a standard limit f_∞ . Transfer of the ordinary convergence $f_{k,3} \rightarrow f_\infty$ then yields ${}^*f_{H,3} \approx_{C^1} {}^*f_\infty$ for every unlimited H . This does not infer nearstandardness from boundedness alone. Because the shadow is in C^1 , standard part commutes with its derivative, and the infinitesimal metric identity becomes exact.

This is Corollary 10 in a different coordinate system of thought. It is not a stronger theorem.

16. Proposition-by-proposition translation

Paper item	Hyperfinite form	Status
Proposition 1	*f_0 satisfies transferred C1-C4 and endpoint equations	exact transfer; the positive margin may be infinitesimal near the seam
(4.8)-(4.11)	phase/Bessel formula accepts internal curves and frequencies	exact transfer
Lemma 2	$N\Delta \in {}^*\mathbb{Z} \Rightarrow$ exact quotient descent	exact; "infinitesimally integral" is insufficient
Lemma 3	transverse error $\leq {}^*C/N$	infinitesimal only if $C/N \approx 0$
Lemma 4	$\nu \in \ker {}^*\ell_i \iff a_p = {}^*\ell_i(p) + c$	exact algebra
Lemmas 5-6	selected primitive defect becomes infinitesimal	one coordinate only; constants must be dominated
Lemma 7	$2\pi N_{k,i} \ell_i(e_X) \in {}^*\mathbb{Z}$	exact seam certificate
Lemma 8	blending error is ${}^*O(1/N_{k,i})$ in C^1	exact gluing plus scale condition
Theorem 9	transferred stage estimates at hyperfinite k	works because displayed C, C' are independent of k
Corollary 10	every unlimited stage has the same C^1 shadow	equivalent reformulation
Theorem 11	transferred one-step bounds	early derivative changes may remain appreciable
Lemma 12	flow field is never exactly horizontal	does not imply an appreciable angle
Theorem 13	finite remainder becomes ordered hyperfinite tail product	shadow exists because the classical product converges

17. Hyperfinite Gauss product

For unlimited H , define

$$\mathcal{R}_H(k, i) = \left(\prod_{\ell=k+1}^H \prod_{j=1}^3 {}^*C_{\ell, j} \right) \left(\prod_{j=i}^3 {}^*C_{k, j} \right).$$

Every finite factor lies in $SO(3)$, so by transfer the internal ordered product lies exactly in ${}^*SO(3)$. For standard (k, i) , convergence of the paper's tail product implies

$$R(k, i) = \text{st } \mathcal{R}_H(k, i),$$

independently of the unlimited cutoff H .

At a hyperpoint infinitesimally close to the cap boundary, the first active index can itself be unlimited. That is the clean NSA picture of the fractal expansion: the cap monad sees only an infinitesimal tail; an ordinary point in the central belt sees a product beginning at a finite stage.

One cannot infer a Loeb distribution or a fractal dimension from this product without new estimates.

Part III - Original research: the two radius regimes

18. Infinitesimal-radius scale-loss theorem

The paper's final theorem can be encoded as an internal statement:

$$(\forall r)(0 < r < 1 \Rightarrow \exists f[f : S^2 \rightarrow B_r \wedge f \in C^1 \wedge I_f = I_h]).$$

The exact coding of C^1 maps, derivatives, and pullback metrics lives in the superstructure and contains no “standard” predicate. Transfer therefore allows a nonstandard radius.

Theorem 18.1 - scale loss NEW-COROLLARY

Let $0 < \varepsilon \approx 0$. There is an internal exact isometric map

$$F_\varepsilon : {}^*S^2 \rightarrow {}^*B_\varepsilon, \quad I_{F_\varepsilon} = {}^*I_h.$$

Then:

1. $\text{st}(F_\varepsilon(p)) = 0$ for every standard p .
2. For every standard C^1 path $\gamma : [0, 1] \rightarrow S^2$, ${}^*L_{\mathbb{R}^3}(F_\varepsilon \circ {}^*\gamma) = {}^*L_{S^2}({}^*\gamma) = {}^*(L_{S^2}(\gamma))$.
3. If $f^b : S^2 \rightarrow \mathbb{R}^3$ is the pointwise shadow $f^b(p) = \text{st } F_\varepsilon(p)$, then $f^b \equiv 0$. Consequently, $\text{st}({}^*L_{\mathbb{R}^3}(F_\varepsilon \circ {}^*\gamma)) = L_{S^2}(\gamma)$, while $L_{\mathbb{R}^3}(f^b \circ \gamma) = 0$. Taking pointwise shadow before forming length changes the answer whenever $L_{S^2}(\gamma) > 0$.
4. In any fixed standard local chart and standard smooth tangent-frame trivialization around a standard point p , the represented derivative field dF_ε is not S-continuous at p .

5. For appreciably separated standard p, q , their ambient separation is infinitesimal while their intrinsic separation is appreciable.

Proof

Transfer gives F_ε and its exact internal metric identity. Ball containment gives $\|F_\varepsilon(p)\| < \varepsilon$, hence every standard point shadows to zero.

Length is the integral of speed. Pulling back the Euclidean metric by an exact isometry preserves the speed of every internal tangent vector, so transfer of the ordinary length identity gives item 2. Taking the standard part of that limited scalar gives the first equality in item 3; applying ordinary Euclidean length to the constant pointwise shadow gives the second.

Suppose the represented dF_ε were S-continuous at a standard p in one such chart and frame. The local standard-part differentiation theorem would identify the derivative of the shadow with the shadow of the represented derivative. The former is zero. The latter sends every standard unit tangent vector to a unit vector because the internal metric identity is exact. Contradiction. Standard smooth changes of frame preserve this conclusion.

Finally,

$$\|F_\varepsilon(p) - F_\varepsilon(q)\| \leq 2\varepsilon \approx 0,$$

while the round distance of appreciably separated points is appreciable. $\square$

Why this is not a contradiction

Internal $*C^1$ continuity permits a nonstandard modulus of continuity. The derivative can change substantially over intervals smaller than every standard interval. The map is S-continuous in position-it is even globally infinitesimally close to zero-but it is not S- C^1 .

Compare the standard-radius hyperfinite stage:

regime	position shadow	derivative shadow	result
standard r , unlimited construction stage	nontrivial	exists continuously	standard C^1 isometry
infinitesimal ε , transferred final theorem	constant zero	cannot form a continuous standard derivative	exact internal isometry with no nontrivial standard shadow

19. No single-scale camera theorem

Proposition 19.1 NEW-COROLLARY

No internal similarity factor $\lambda > 0$ makes both of the following appreciable and limited for λF_ε :

- the global diameter;
- the image speed of a standard unit tangent vector.

Proof

The global diameter is at most $2\lambda\epsilon$. If λ is limited, this is infinitesimal. If the global diameter is appreciable, then $\lambda\epsilon$ is appreciable, so λ is unlimited. Exact isometry gives

$$\|d(\lambda F_\epsilon)(v)\| = \lambda\|v\|,$$

which is unlimited for a standard unit v . $\square$

This is the formal version of the product decision:

- a global gauge makes the collapsed ball visible;
- an intrinsic gauge makes walking human-sized;
- a microscope gauge makes one corrugation scale visible.

They can be synchronized by address, but not identified as one Euclidean camera scale.

20. Hyperfinite path length and resolution

Let $F : {}^*S^2 \rightarrow {}^*\mathbb{R}^3$ be an internal ${}^*C^1$ map with exact pullback metric $I_F = {}^*I_h$, let $\gamma : [0, 1] \rightarrow S^2$ be a standard C^1 , finite-length path, and set $g = F \circ {}^*\gamma$. Choose a positive infinitesimal α . Because g' is internal and continuous on ${}^*[0, 1]$, transferred uniform continuity supplies an internal $\delta > 0$ such that

$$|s - t| < \delta \implies \|g'(s) - g'(t)\| < \alpha.$$

Choose an internal hyperfinite partition

$$0 = t_0 < t_1 < \dots < t_M = 1, \quad M \in {}^*\mathbb{N} \setminus \mathbb{N}.$$

whose mesh is smaller than δ . Its polygonal sum differs infinitesimally from the internal speed integral. Therefore, for an exact isometry,

$$L_{S^2}(\gamma) = \text{st} \sum_{j=0}^{M-1} \|F({}^*\gamma(t_{j+1})) - F({}^*\gamma(t_j))\|.$$

" M is infinite" is not by itself a resolution certificate; the actual criterion is the derivative modulus $\omega_{g'}(1/M) \approx 0$. In a particular cutoff construction, $M/N_{\max}$ being unlimited is a useful frequency heuristic, but it is sufficient only after the phase derivatives, transition cutoffs, compositions, and amplitudes have been included in a proof of that modulus. Conversely, a high-frequency component with infinitesimal amplitude need not force that ratio to be unlimited.

For a micro-isometric stage with relative metric error $\eta \approx 0$, a limited path length L has infinitesimal accumulated length error. For an unlimited walk, require the stronger condition

$$\eta L \approx 0.$$

This is the mathematical reason simulator stress can validate state stability without certifying an arbitrarily long physical path.

21. The actual “infinite distances problem”

The compact source sphere satisfies

$$\text{diam}(S^2, d_{S^2}) = \pi.$$

No pair of points has infinite minimal geodesic distance. Three other quantities can be unlimited:

- 1. the ratio of appreciable intrinsic distance to infinitesimal ambient chord;
- 2. the length of a path that loops an unlimited number of times;
- 3. the sampling frequency required to resolve an internal derivative field.

These must not be conflated.

Part IV - App semantics derived from the mathematics

22. Intrinsic address is the identity

APP-MODEL The authoritative user state is

$$(q, a, \psi), \quad q \in S^2,$$

where q is a source-sphere point, a is signed altitude, and ψ is heading. The reduced-space point is an observation of q , not its identity.

A tangent step $v \in T_q S^2$ of length $s = \|v\|$ updates the address by

$$q' = \cos(s)q + \sin(s)\frac{v}{s}.$$

This exponential-map update has no longitude singularity. The visual world re-anchors around q , keeping RealityKit coordinates bounded while the accumulated intrinsic path length grows.

23. Four regimes, four invariants

regime	invariant ruler	what moves	mathematical reading
Habitat	intrinsic meters per unit-sphere radian	world re-anchors under user	live on the parameter sheet
Hover	intrinsic address plus signed normal altitude	same address, bounded vertical gauge	live above the sheet
Interior	containing-ball radius	shell surrounds observer	inspect extrinsic collapse from within

regime	invariant ruler	what moves	mathematical reading
Atlas	finite whole-object fit	sphere rotates/scales as an exhibit	compare global ambient and intrinsic scales

Changing regime is a gauge transition. The UI should not animate it as if one fixed-scale camera traversed the implied mathematical distance.

24. Collision must be sheet-aware

The reduced image can place intrinsically distant points extremely close in $\mathbb{R}^3$. A nearest-triangle query in ambient space can therefore jump to the wrong source point.

APP-MODEL Each rendered patch and collision sample retains:

```
intrinsic address
stage and ribbon identity
local tangent frame
extrinsic position
mesh-resolution receipt
```

Walking follows adjacency in the intrinsic triangulation. Ambient collisions may prevent visual penetration, but they never choose the next address.

25. What the public app can honestly say

It can say:

- “paper-calibrated finite proxy”;
- “the smooth-cap / corrugated-belt decomposition follows the paper”;
- “this address advances by the unit-sphere metric”;
- “this simulator run exercised the rendered finite mesh”;
- “the NSA scale-loss theorem motivates the gauge transition.”

It cannot say:

- that the runtime proxy is the authors' $f_{1,3}$;
- that a finite mesh is isometric;
- that the exact limit has been sampled;
- that a global pointwise shadow preserves intrinsic geometry;
- that visual smoothness proves headset comfort; or
- that the C^1 -fractal has a computed Hausdorff dimension.

Part V - Proof obligations and formalization hooks

26. Fifteen fail-closed obligations

1. **Foundation:** choose enlargement or IST and translate explicitly.
2. **Internality:** never transfer a formula containing $\approx$, "limited," or "standard."
3. **Finite atlas:** internalize the sphere and derivative norms with a fixed standard atlas.
4. **Hidden constants:** expand every $O(1/N_{k,i})$ to $C_{k,i}/N_{k,i}$.
5. **Boundary margins:** a positive primitive defect can be infinitesimal near a seam.
6. **Arithmetic closure:** quotient congruences remain exact.
7. **Target convergence:** control the boundary strips uniformly, not only each fixed compact subribbon.
8. **Nearstandardness:** boundedness in C^1 is not enough in an infinite-dimensional space.
9. **Derivative shadow:** commute standard part and d only after proving C^1 -nearstandardness.
10. **Exact versus micro:** distinguish the transferred exact final map from an unlimited finite-stage micro-isometry.
11. **Quadrature:** dominate active frequency with sampling frequency.
12. **Long walks:** bound accumulated ηL .
13. **Sheet identity:** never infer intrinsic adjacency from ambient nearest neighbors.
14. **Injectivity:** preserve the paper's wording distinction until a global injectivity lemma is supplied.
15. **Fractal language:** no dimension or curvature blow-up claim without new estimates.

27. A Lean-shaped decomposition

The shortest path to formal confidence is not to formalize a nonstandard universe first. Formalize the ordinary convergence skeleton, then prove that the NSA statement is an equivalence in whichever NSA library is selected.

Layer A - ordinary analysis

```
StageData:
  map f_k
  target metric I_k
  active ribbons D_k_i
  derivativeIncrementBound
  metricErrorBound
  exactBoundaryAgreement

StageContract k:
  metricError f_k <= C * sqrt(tau (k-1))
  dist (derivative f_k) (derivative f_(k-1)) <= C' * tau (k-1)

ConvergenceTarget:
  Summable tau
  CompleteSpace C1Map
  pullbackMetric continuous in C1
```

Target lemmas:

```
summable_stage_derivatives
c1_cauchy_of_stage_contract
boundary_agreement_closed_under_c1_limit
pullback_metric_tendsto
isometric_limit
ball_containment_of_uniform_clearance
```

Layer B - geometry and paths

```
SphereAddress := { q : R3 // norm q = 1 }
expStep : SphereAddress -> Tangent q -> SphereAddress

theorem expStep_norm
theorem expStep_zero
theorem expStep_distance_small
theorem intrinsic_path_length_invariant
```

Layer C - NSA bridge

The theorem target is not “take standard part” as a magical operation. It is:

```
hyperCauchy F :=
  for all infinite H K, dist (star F H) (star F K) is infinitesimal

theorem c1_limit_iff_hyperCauchy
theorem microMetric_of_infinite_stage
theorem shadowMetric_eq
```

The infinitesimal-radius result is a separate theorem with no derivative shadow:

```
theorem scale_loss
  (epsilon_pos : 0 < epsilon)
  (epsilon_inf : Infinitesimal epsilon)
  (F : InternalIsometry sphere (ball epsilon)) :
  pointwiseShadow F = const 0 /\
  preservesInternalLength F /\
  not (SContinuous (derivative F) atEveryStandardPoint)
```

Treat these as specification signatures. The exact syntax depends on the NSA foundation and manifold library.

28. A practical proof order for Alok

1. Prove the no-single-scale proposition in plain ordered-field algebra.
2. Prove the unit-sphere exponential-map update preserves norm.
3. Formalize a generic Banach-space theorem: summable increments imply a Cauchy sequence and a limit.
4. Instantiate it with abstract `C1Map` and an assumed continuous pullback metric operator.
5. Only then encode the paper-specific Stage Contract.
6. Add the NSA equivalence as a reflection theorem about unlimited indices.
7. Keep the transferred infinitesimal-radius theorem in a separate namespace; its failure of C^1 shadow is a feature, not a failed proof.

Part VI - New research directions

29. Questions opened by the scale-loss theorem

29.1 Minimal multiscale atlas OPEN

What is the least family of blow-up gauges $\{\sigma_j\}$ needed to recover the frame product up to a given standard angular tolerance? This is more precise than asking for a single level-of-detail mesh.

29.2 Loeb distribution of normals OPEN

At an unlimited stage, push a hyperfinite counting measure on a resolved grid through the Gauss map and take its Loeb measure. Does the boundary activation index induce a measurable decomposition whose standard moments match finite normal-scale diagnostics? No such result follows from the paper alone.

29.3 Conditioned hyperfinite frequency selection **OPEN**

Can one replace the recursively selected standard frequency schedule by one internal hyperfinite choice whose exact quotient congruences, primitive margins, and all $C_{k,i}/N_{k,i}$ constraints hold simultaneously? This is a natural idealization/saturation problem, but the stage-dependent constants must be internalized first.

29.4 Intrinsic collision semantics **[OPEN / APP-RESEARCH]**

How should an immersive engine represent two source-sheet points that are ambiently indistinguishable at the active pixel/depth tolerance? A promising answer is a typed contact relation indexed by intrinsic patch identity, with ambient contact as evidence rather than identity.

29.5 Curvature microscope **OPEN**

The derivative must lose S-continuity in the infinitesimal-radius regime, but this report does not prove a quantitative second-derivative or curvature blow-up law. Deriving one would require explicit frequency/amplitude estimates beyond the displayed convergence theorems.

30. Source and provenance ledger

Reduced sphere

- Bartzos, Borrelli, Denis, Lazarus, Rohmer, Thibert, "An Explicit Isometric Reduction of the Unit Sphere into an Arbitrarily Small Ball."
- Project PDF: <https://hevea-project.fr/pdfSphere/focm-revised-2.pdf>
- DOI: <https://doi.org/10.1007/s10208-017-9360-1>
- HAL author deposit: <https://inria.hal.science/hal-01647062v1>
- Locally verified project-PDF SHA-256: `5f8588a257a173ef86b8ed45bdabba05f09428c6229e6f4055a50ccec2e2a141`
- Official gallery: https://hevea-project.fr/Sphere_Gallery.html
- Gallery HTML SHA-256: `8ca228e881225d294115386c3438ef92edc01c0ef333911888f2e4f583ea7d4f`
- 78 high-resolution stills were inventoried.
- Images/videos are CC BY-SA 2.0 France and credited to E. Bartzos, V. Borrelli, R. Denis, F. Lazarus, D. Rohmer, and B. Thibert.
- The attached derivative's SHA-256 is `4821cac3bc0aeb8a31f3518c718c5b4a59191d7048b7418c38d5b1b265edd9c0`.
- Official meshes/code page: https://hevea-project.fr/Hevea_3Dprint_code.html
- The sphere WRL downloads have no explicit adjacent mesh license.
- They were inspected as references and are not redistributed.
- Official GPL flat-torus source: <https://github.com/HeveaProject/Hevea>
- pinned revision `e792074e4dd6319351bc957afeb16b4725d304f0` ;
- it contains the flat-torus generator, not the reduced-sphere generator.
- Community GPL reproduction: <https://github.com/Juddd/hevea-reduced-sphere>

- inspected revision [c098ea6fabb994bdd2555719b64ebbe8d7fca483](#) ;
- it independently fits a degree-thirteen profile to public mesh envelopes and implements the paper's flow architecture;
- it explicitly does not claim recovery of the authors' hidden degree-nine coefficients.
- Live author clarification: <https://github.com/HeveaProject/Hevea/issues/1#issuecomment-5396587822>
- Roland Denis posted the composite-Hermite profile recipe and the article parameters `theta = 88 degrees` , `eta = 0.5` , and `beta = 0.6` on 2026-08-24, while saying that transition-function details would follow;
- the comment's printed cap/belt interval formulas use `pi / 2 - theta` , which is inconsistent with `theta = 88 degrees` , the paper's condition (3.4), the public mesh envelope, and the reproduced `yInfinity` near 88 degrees;
- until that convention is clarified, this report treats the comment as valuable live correspondence, not as an unambiguous executable baseline.

Nonstandard analysis

- Edward Nelson, "Internal Set Theory: A New Approach to Nonstandard Analysis," *Bulletin of the AMS* 83 (1977), 1165-1198.
- Author PDF: <https://web.math.princeton.edu/~nelson/papers/ist.pdf>
- DOI: <https://doi.org/10.1090/S0002-9904-1977-14398-X>
- Transfer/Idealization/Standardization: pp. 1166-1169.
- Standard part: p. 1169.
- Infinitesimal differentiation and hyperfinite subdivisions: pp. 1181-1184.
- Nearstandard points and compactness: pp. 1185-1186.
- Abraham Robinson, "Non-standard Analysis," *Indagationes Mathematicae* 23 (1961), 432-440.
- DOI: [https://doi.org/10.1016/S1385-7258\(61\)50044-3](https://doi.org/10.1016/S1385-7258(61)50044-3)
- H. Jerome Keisler, *Foundations of Infinitesimal Calculus*.
- Author edition: <https://people.math.wisc.edu/~hkeisler/foundations.pdf>
- Standard part, infinite-index convergence, hyperfinite line integrals, internal/S-continuity, overspill, and saturation are the relevant chapters.
- Tahl Nowik and Mikhail G. Katz, "Differential Geometry via Infinitesimal Displacements."
- Primary paper: <https://arxiv.org/pdf/1405.0984>
- Relevant for internal maps on manifolds, compact nearstandardness, and the scope of saturation.

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32. Final compression

The paper's classical proof and the NSA view align cleanly at a standard radius:

summable stage errors $\iff$ one unlimited stage with infinitesimal C^1 tail $\implies$ standard C^1 isometry.

The genuinely new phenomenon appears only after transferring the finished theorem to an infinitesimal radius:

exact internal metric + infinitesimal global image $\implies$ constant pointwise shadow but nonzero internal lengths.

So the correct immersive representation is not one mesh at one scale. It is a commuting system of views indexed by one intrinsic address:

```
intrinsic source address
-> habitat blow-up
-> hover gauge
-> interior gauge
-> global atlas gauge
```

The address is the geometry. The rendered point is only the current observation of it.